Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Introduction

In modern organizations, managing employee data efficiently is crucial for smooth business operations. Manual data entry often leads to inconsistencies, duplication, and errors. This project focuses on automating the process of importing employee data into ServiceNow and establishing relationships between records using Import Sets, Transform Maps, and Dot Walking.

Objective

The main objective of this project is to automate data import processes, ensure accurate relationship mapping, reduce manual effort, and improve data accessibility using ServiceNow features.

Tools and Technologies Used

- ServiceNow Platform - Google Sheets - CSV File Format - Import Sets - Transform Maps - Dot Walking

Detailed Methodology

1. Created employee dataset in Google Sheets with fields like name, email, department, manager, and phone number.

2. Exported the dataset as a CSV file.

3. Uploaded the file into ServiceNow using the Load Data module.

4. Created an Import Set table to temporarily store the data.

5. Configured a Transform Map to map source data to the target table (sys_user).

6. Used Mapping Assist to align fields correctly.

7. Handled reference fields such as department and manager to maintain relationships.

Implementation with Screenshots

Step 1: Uploading employee data and creating import set table.

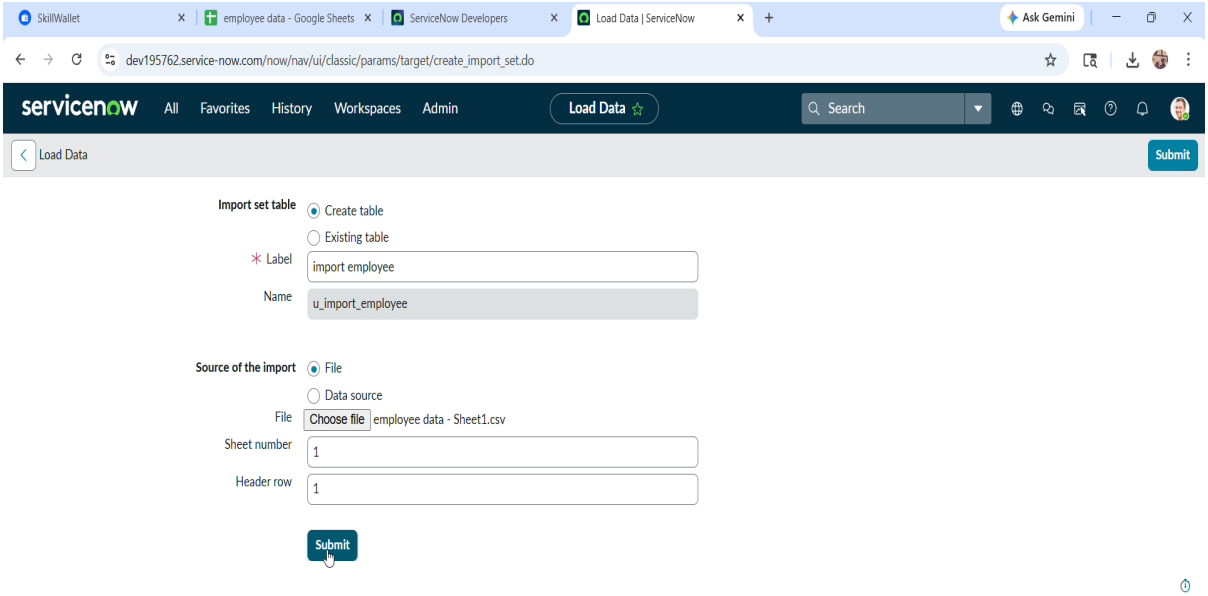

Fig 1: Import Set Creation and Data Upload

Step 2: Creating Transform Map to map source and target tables.

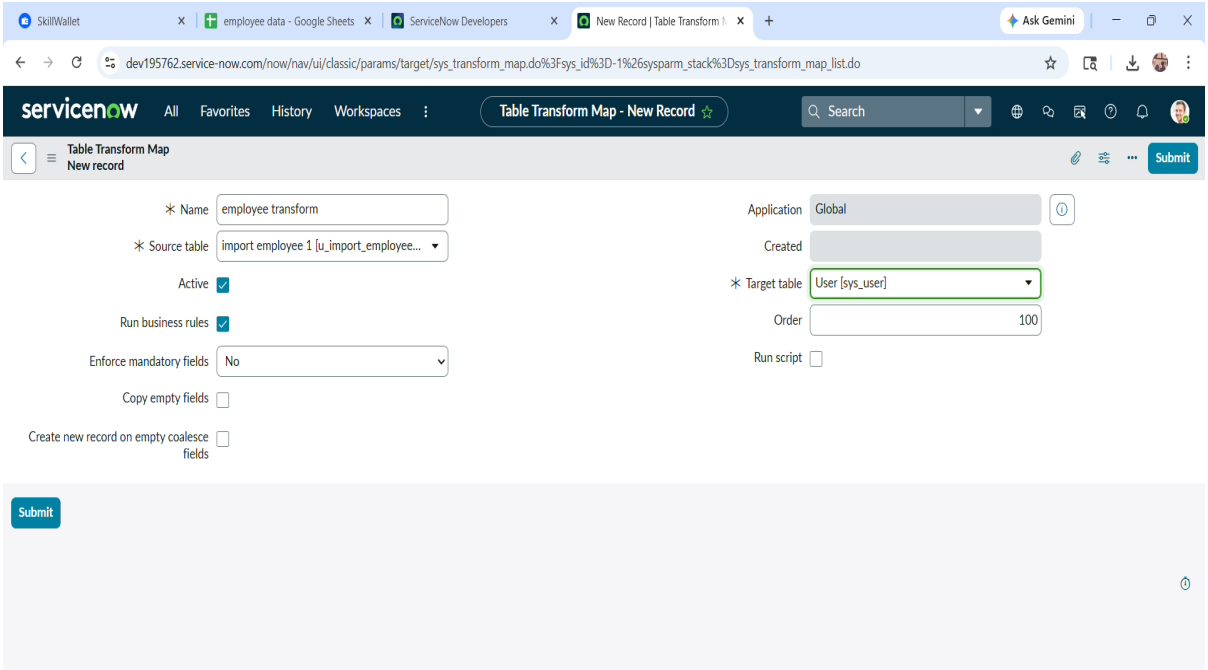

Fig 2: Transform Map Configuration

Step 3: Mapping fields using Mapping Assist.

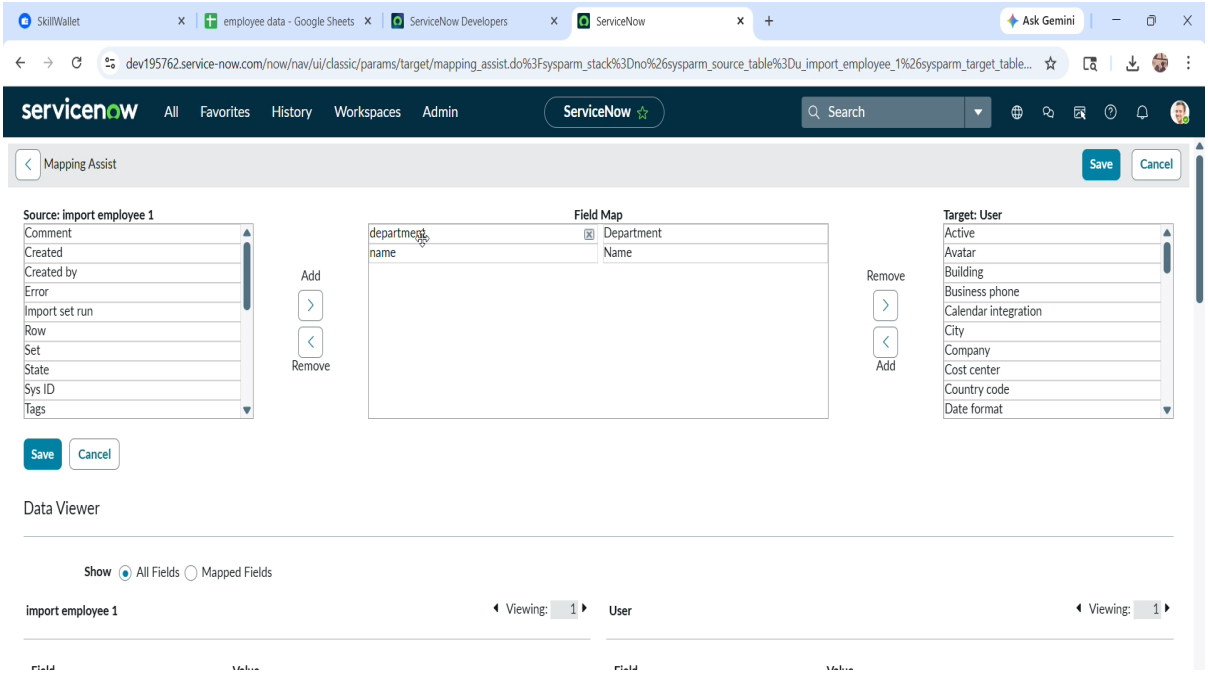

Fig 3: Field Mapping using Mapping Assist

Step 4: Configuring Incident form and implementing Dot Walking.

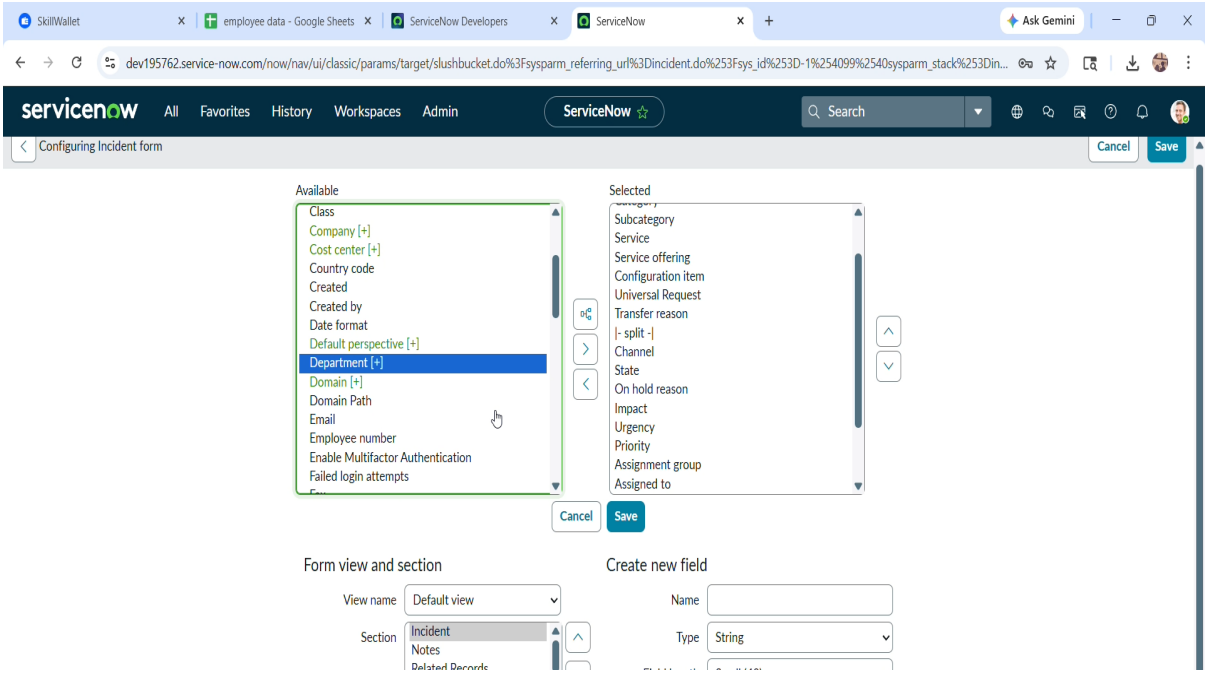

Fig 4: Dot Walking Configuration

Results and Benefits

The project successfully automated the data import process, reducing manual effort and minimizing errors. It ensured accurate data mapping and improved data consistency. Dot walking enabled dynamic access to related data without redundancy, enhancing user experience.

Challenges Faced

- Handling reference fields correctly (department and manager)

- Ensuring accurate field mapping

- Avoiding duplicate records during transformation

Future Enhancements

- Automating scheduled data imports

- Integrating with external HR systems

- Adding validation rules for better data quality

Conclusion

This project demonstrates how automation in ServiceNow can significantly improve data management processes. By using Import Sets, Transform Maps, and Dot Walking, organizations can ensure efficient, accurate, and scalable data handling systems.